

1-Bit Compressive Sensing

Zachary Dorson

Tufts University

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Compressive Sensing

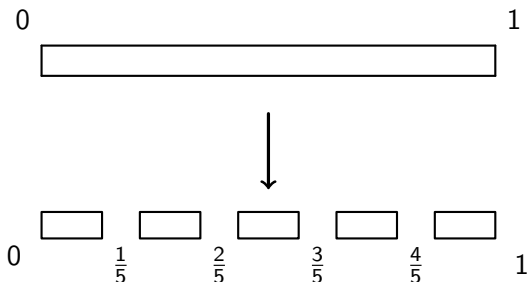
Brief overview

- k -sparse signal $\mathbf{x} \in \mathbb{R}^n$
- M linear measurements $\mathbf{y} = \Phi\mathbf{x}$, $\Phi \in \mathbb{R}^{M \times n}$ with rows φ_i
- Want to find:

$$\min_{\hat{\mathbf{x}}} \|\hat{\mathbf{x}}\|_0 \quad \text{s.t.} \quad \Phi\hat{\mathbf{x}} = \mathbf{y}$$

Quantization

- In practice our measurements must be stored.
- Both the measurement process and storage introduce precision loss
- This is called **quantization**

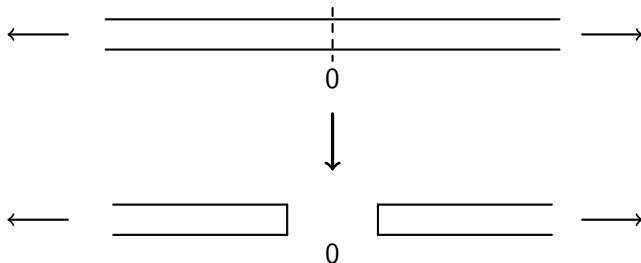


1-Bit Quantization

- We will consider the most extreme form of quantization, where we keep only the sign

$$y = \text{sign}(\Phi x)$$

- Each measurement stored with a single bit



Why 1-bit

When we'd rather take many low-resolution measurements than fewer high-resolution ones:

- Radar systems
- Antenna arrays

Key Consideration

- Sign measurements carry no magnitude information, so we can only recover directions
- So we search for solutions on the unit sphere
- Can think of this geometrically: each measurement $\text{sign}(\varphi_i^T \mathbf{x})$ tells us which side of a hyperplane $\mathbf{x}$ lies on

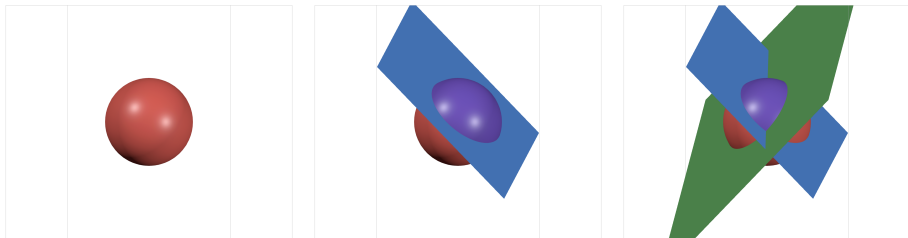


Figure: Planes sectioning the sphere. Each measurement shrinks the shaded region

Reformulate the sign constraints:

$$\text{sign}(\langle \phi_i, \hat{\mathbf{x}} \rangle) = y_i \implies \text{diag}(y) \Phi \hat{\mathbf{x}} \geq 0$$

Then we take the problem

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{x}\|_1 \quad \text{s.t.} \quad \text{diag}(y) \Phi \mathbf{x} \geq 0, \quad \|\mathbf{x}\|_2 = 1$$

and relax the sign consistency constraint to

$$\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{x}\|_1 + \lambda \sum_i f((\text{diag}(y) \Phi \mathbf{x})_i), \quad \|\mathbf{x}\|_2 = 1$$

with cost function $f(x) = \frac{x^2}{2}$ when $x < 0$ and $f(x) = 0$ else.

RFPI (Boufounos & Baraniuk)

Sample initial iterate from the unit sphere $\hat{\mathbf{x}}_0 \sim S^{n-1}$, set step size δ and parameter λ . Then

Algorithm 1 Renormalized fixed point iteration (RFPI)

```
1: for  $t = 1, 2, 3, \dots$  until convergence do  
2:    $\bar{\mathbf{f}}_t \leftarrow (\mathbf{Y}\Phi)^T \bar{f}'(\mathbf{Y}\Phi\hat{\mathbf{x}}_{t-1})$   
3:    $\tilde{\mathbf{f}}_t \leftarrow \bar{\mathbf{f}}_t - \langle \bar{\mathbf{f}}_t, \hat{\mathbf{x}}_{t-1} \rangle \hat{\mathbf{x}}_{t-1}$   
4:    $\mathbf{h} \leftarrow \hat{\mathbf{x}}_{t-1} - \delta \tilde{\mathbf{f}}_t$   
5:    $\mathbf{u} \leftarrow \text{sign}(\mathbf{h}) \max \{ |\mathbf{h}| - \frac{\delta}{\lambda}, 0 \}$   
6:    $\hat{\mathbf{x}}_t \leftarrow \frac{\mathbf{u}}{\|\mathbf{u}\|_2}$   
7: end for
```

- Problem: we need λ sufficiently large, but not too large or the soft thresholding is ineffective. But we cannot determine the right λ in advance.

Fixing The Problem

- Wrap algorithm 1 in an outer loop that starts each iteration at the previous solution
- Increase λ each iteration, until the outer loop converges

Algorithm 2 RFPI with continuation

- 1: Set continuation parameters λ_0, β
 - 2: $\hat{\mathbf{x}}_0 \sim S^{n-1}$
 - 3: **for** $i = 1, 2, 3, \dots$ until convergence **do**
 - 4: $\lambda_i = \lambda_0 \cdot \beta^{i-1}$
 - 5: $\hat{\mathbf{x}}^{(i)} \leftarrow$ result of RFPI with initial iterate $\hat{\mathbf{x}}^{(i-1)}$ and $\lambda = \lambda_i$
 - 6: **end for**
-

- Simpler than RFPI, no continuation schedule.
- Uses a different penalty.

Let $\Sigma_k^n \subset \mathbb{R}^n$ denote the set of k -sparse vectors and $\eta = \sqrt{2\pi}$. Then

Algorithm 3 Binary iterative hard thresholding (BIHT)

```
1:  $\hat{\mathbf{x}}^{(0)} \sim S^{n-1} \cap \Sigma_k^n$ 
2: for  $t = 1, 2, 3, \dots$  do
3:    $\tilde{\mathbf{x}}^{(t)} \leftarrow \hat{\mathbf{x}}^{(t-1)} + \frac{\eta}{M} \Phi^T \cdot \frac{1}{2} (\mathbf{y} - \text{sign}(\Phi \hat{\mathbf{x}}^{(t-1)}))$ 
4:    $\tilde{\mathbf{x}}^{(t)} \leftarrow \text{Proj}_{\Sigma_k^n}(\tilde{\mathbf{x}}^{(t)})$ 
5:    $\hat{\mathbf{x}}^{(t)} \leftarrow \frac{\tilde{\mathbf{x}}^{(t)}}{\|\tilde{\mathbf{x}}^{(t)}\|_2}$ 
6: end for
```

Convergence Guarantees

- RFPI: No guarantee we converge to global minimum
- BIHT: Shown that we do converge to the true solution

Theorem (informal)

Let $x \in \mathcal{S}^{n-1}$ be k -sparse, and let $\Phi \in \mathbb{R}^{M \times n}$ have i.i.d. Gaussian rows. For any $\epsilon > 0$, if

$$M = O\left(\frac{k}{\epsilon} \log \frac{n}{k\sqrt{\epsilon}}\right),$$

then with high probability the BIHT iterates satisfy

$$d_{\mathcal{S}^{n-1}}(x, \hat{x}^{(t)}) \leq 2^{2^{-t}} \epsilon^{1-2^{-t}}, \quad t = 0, 1, 2, \dots$$

- Proof relies on an RIP-like property, the restricted approximate invertibility condition (RAIC)

Reconstruction Performance

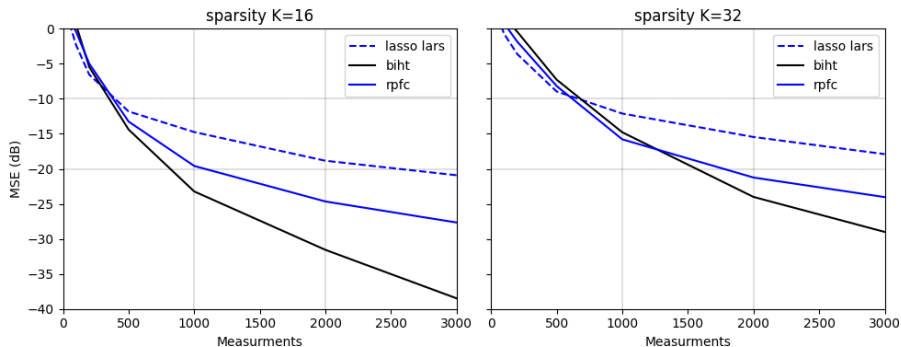


Figure: Reconstruction performance for baseline, RFPI*, BIHT

More on RFPI

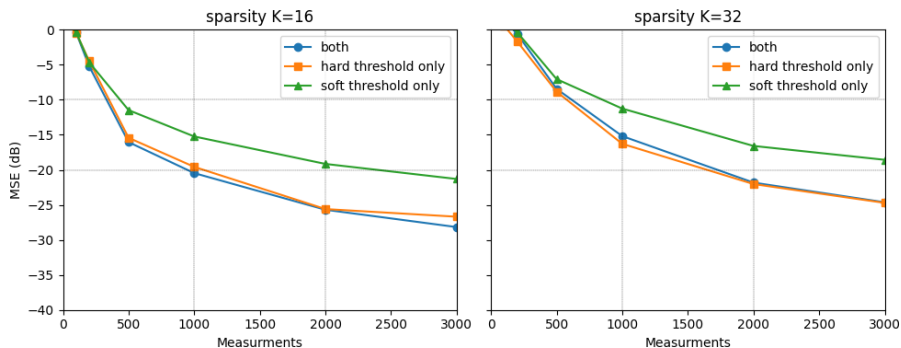


Figure: Performance of variations of RFPI

2018 Survey by Li et al. notes uses of 1-bit specific algorithms in:

- Wireless sensor networks
 - Constraints on bandwidth and energy
- Radar
 - Signals encoded with few bits to save cost